

STT3A-Congenital Disorder of Glycosylation (STT3A-CDG): Comprehensive Disease Characterization Report

Disease: STT3A-Congenital Disorder of Glycosylation (CDG type Iw) **MONDO ID:** MONDO:0014270 **OMIM:** #615596 (autosomal recessive) and #619714 (autosomal dominant) **Gene:** *STT3A* (HGNC:30591; NCBI Gene 3703; chr11q24.1; UniProt P46977; OMIM 601134) **Category:*** Mendelian inborn error of metabolism

Summary

STT3A-CDG is an ultra-rare, multisystem inborn error of protein N-glycosylation caused by pathogenic variants in *STT3A*, the gene encoding the catalytic subunit of the co-translational oligosaccharyltransferase complex (OST-A). The OST-A complex, positioned adjacent to the Sec61 translocon in the endoplasmic reticulum (ER) membrane, transfers a pre-assembled Glc₃Man₉GlcNAc₂ oligosaccharide from a dolichol-pyrophosphate donor onto asparagine residues within N-X-S/T sequons of nascent polypeptides as they emerge into the ER. When OST-A activity is reduced, a subset of glycoproteins — particularly those whose sequons are glycosylated preferentially by STT3A during translation — become hypoglycosylated and destabilized, producing a **CDG type I** biochemical signature (an abnormal serum transferrin pattern with increased di- and a-sialotransferrin).

Clinically, STT3A-CDG is a predominantly neurodevelopmental and neuromuscular-skeletal disorder. The autosomal recessive form (first described in 2013) features seizures, developmental delay, intellectual disability, hypotonia, feeding problems/failure to thrive, and a type I transferrin pattern; some patients also have a coagulopathy (low Factor VIII and von Willebrand Factor). A distinct autosomal dominant form (defined in 2021 from 16 individuals across 9 families) is caused by heterozygous missense variants clustering in the catalytic active site and presents with skeletal anomalies, short stature, macrocephaly, dysmorphism,

increased muscle tone/cramps, and intellectual disability in about half of cases. The two inheritance mechanisms — biallelic loss-of-function versus heterozygous dominant-negative active-site variants — represent one of the more unusual features of this disorder, reflected in dual OMIM entries.

Diagnosis rests on transferrin isoelectric focusing (or HPLC/capillary electrophoresis) followed by confirmatory exome/genome or targeted *STT3A* sequencing. There is no disease-specific or curative therapy; unlike certain treatable CDG subtypes (PGM1-CDG with galactose, MPI-CDG with mannose, TMEM165-CDG with galactose), the dietary sugar therapies do not apply to STT3A-CDG because it is an ER glycan-transfer defect rather than a Golgi processing defect. Management is supportive and multidisciplinary. Disease mechanism has been validated across yeast (*S. cerevisiae* STT3), zebrafish (CRISPR heterozygous knockdown), and human patient cell models, and gnomAD population-constraint metrics quantitatively corroborate that *STT3A* is strongly intolerant to both missense and loss-of-function variation.

Key Findings

Finding 1 — STT3A-CDG is caused by defects in the catalytic subunit of the OST-A complex

STT3A (chromosome 11q24.1; OMIM 601134) encodes the catalytic subunit of the STT3A-containing oligosaccharyltransferase (OST-A) complex, which co-translationally transfers the $\text{Glc}_3\text{Man}_9\text{GlcNAc}_2$ glycan from dolichol-pyrophosphate onto asparagine residues within N-X-S/T sequons. The original 2013 report by Shrimal, Ng, Losfeld, Gilmore, and Freeze identified a homozygous *STT3A** c.1877T>C variant (p.Val626Ala) causing a recessive CDG; the mutation impaired glycosylation of a GFP-based biomarker and was rescued by wild-type cDNA.

"which are caused by mutations in different isoforms of the catalytic subunit of the oligosaccharyltransferase (OST)" — [PMID: 23842455](#)

"A homozygous mutation (c.1877T > C) in *STT3A* causes a p.Val626Ala change" — [PMID: 23842455](#)

"*STT3A* encodes the catalytic subunit of the STT3A-containing oligosaccharyltransferase (OST) complex, essential for protein N-glycosylation" — [PMID: 34653363](#)

This establishes the fundamental molecular lesion: a defect in the enzyme that initiates the N-glycosylation of nascent proteins in the ER.

Finding 2 — STT3A-CDG has both autosomal recessive and autosomal dominant forms

The recessive form (OMIM #615596, CDG-Iw) was first described in 2013 as an autosomal-recessive disorder with seizures, developmental delay, intellectual disability, and a type I transferrin pattern. In 2021, Wilson et al. reported 16 individuals from 9 families with inherited or de novo heterozygous missense variants clustering in the STT3A catalytic active site, defining a distinct autosomal-dominant CDG (OMIM #619714). Subsequent reports (2024–2026) confirmed and expanded the dominant spectrum, including a zebrafish-validated dominant missense variant (p.Asp167Tyr).

"we here report the identification of 16 individuals from nine families who have either inherited or de novo heterozygous missense variants in STT3A, leading to an autosomal-dominant CDG" — [PMID: 34653363](#)

"all variants are located in the catalytic site of STT3A" — [PMID: 34653363](#)

"Previous studies have reported that STT3A-CDG is caused by autosomal recessive inheritance. However, in this study, we propose that STT3A-CDG can be pathogenic through autosomal dominant inheritance" — [PMID: 39891251](#)

The clustering of dominant variants in the catalytic site supports a **dominant-negative** mechanism, whereby a mutant catalytic subunit incorporated into the OST-A complex poisons its activity.

Finding 3 — Core clinical phenotype: neurodevelopmental impairment, seizures, and a type I transferrin pattern

Recessive STT3A-CDG is characterized by seizures, developmental delay, intellectual disability, hypotonia, failure to thrive/feeding problems, and a type I carbohydrate-deficient transferrin (CDT) pattern. A patient carrying p.Tyr360Ser additionally exhibited chronically low Factor VIII and von Willebrand Factor with hypoglycosylated vWF. The dominant form presents with variable skeletal anomalies, short stature, macrocephaly, dysmorphic features, intellectual disability in approximately 50%, and increased muscle tone with muscle cramps.

"characterized by seizures, developmental delay, intellectual disability, and a type I carbohydrate deficient transferrin pattern" — [PMID: 30701557](#)

"chronically low Factor VIII (FVIII) and von Willebrand Factor (vWF) levels and activities" — [PMID: 30701557](#)

"Affected individuals presented with variable skeletal anomalies, short stature, macrocephaly, and dysmorphic features; half had intellectual disability. Additional features included increased muscle tone and muscle cramps" — [PMID: 34653363](#)

Finding 4 — Mechanism: STT3A mediates co-translational N-glycosylation coupled to the ER translocon

The STT3A OST complex is localized adjacent to the Sec61 protein translocation channel and catalyzes co-translational N-glycosylation using an N-terminal-to-C-terminal scanning mechanism. Accessory subunits DC2 (OSTC) and KCP2 mediate the STT3A–translocon interaction. The paralogous STT3B complex (containing the oxidoreductase MagT1) glycosylates sites skipped by STT3A, including cysteine-proximal sequons, providing partial functional redundancy that likely mitigates the severity of STT3A loss.

"The STT3A complex interacts directly with the protein translocation channel to mediate glycosylation of proteins using an N-terminal-to-C-terminal scanning mechanism" — [PMID: 31433728](#)

"the STT3A isoform of the oligosaccharyltransferase is localized adjacent to the protein translocation channel to catalyze co-translational N-linked glycosylation of proteins in the endoplasmic reticulum" — [PMID: 28860277](#)

"the role of the STT3B complex in mediating cotranslational or posttranslational glycosylation of acceptor sites that have been skipped by the STT3A complex" — [PMID: 25460543](#)

Finding 5 — Ultra-rare with two OMIM entries; part of the broader CDG group

Two OMIM phenotype entries exist: CDG type Iw autosomal recessive (#615596) and autosomal dominant (#619714). Only ~6 recessive cases were reported by 2019 (all p.Val626Ala except one p.Tyr360Ser); the dominant cohort added 16 individuals from 9 families in 2021, with further single/small reports since. No formal prevalence or incidence estimate exists for STT3A-CDG specifically. For context, CDG comprise ~200 disorders, of which PMM2-CDG is by far the most common (~60% of all CDG; incidence ~1 in 33,576 in North America/Europe).

"Autosomal dominant congenital disorder of glycosylation (CDG) type Iw (OMIM# 619714)" — [PMID: 39435313](#)

"STT3A-CDG (OMIM# 615596) is an autosomal recessive N-linked glycosylation disorder" — [PMID: 30701557](#)

"Phosphomannomutase 2-congenital disorder of glycosylation (PMM2-CDG) accounts for about 60 % of all CDGs" — [PMID: 40737785](#)

Finding 6 — CDG natural history: multisystem disease with early-childhood mortality risk

In a Frontiers in CDG Consortium (FCDGC) natural-history cohort of 37 deceased CDG patients, all presented with multisystem features including neurological involvement; the majority died within the first three years of life, and about one-third died of cardiopulmonary failure (others from neurological progression, sepsis, or respiratory infection). CDG frequently cause coagulation abnormalities (antithrombin, protein C, factor XI deficiencies) with both bleeding and thrombosis risk. CDG prevalence in childhood epilepsy cohorts is notable (~4.4%), and transferrin isoelectric focusing is a recommended screen.

"The majority of patients involved in this study died during the first three years of life" — [PMID: 39923392](#)

"All of the patients presented with multisystem features with involvement of the neurological system" — [PMID: 39923392](#)

These mortality data are drawn from the broader CDG group; STT3A-CDG-specific mortality figures are not established.

Finding 7 — Functional models recapitulate the glycosylation defect

In *S. cerevisiae*, expressing STT3 carrying variants homologous to patient alleles induced defective glycosylation of carboxypeptidase Y in a wild-type strain and worsened the defect in the hypomorphic *stt3-7* strain, supporting a dominant pathomechanism. CRISPR-Cas9 heterozygous-knockdown zebrafish modeling p.Asp167Tyr reproduced patient-like phenotypes: craniofacial dysmorphology (increased eye distance, altered ceratohyal angle), reduced mineralized bone, and abnormal locomotion/developmental delay. Patient fibroblasts and HeLa/HEK293T systems show hypoglycosylation of STT3A-specific reporters that is rescued by wild-type *STT3A* cDNA.

"expression of STT3 containing variants homologous to those in affected individuals induced defective glycosylation of carboxypeptidase Y in a wild-type yeast strain" — [PMID: 34653363](#)

"Heterozygous knockdown zebrafish exhibit phenotypes similar to those of patients, including craniofacial dysmorphology" — [PMID: 39891251](#)

Finding 8 — Downstream consequence: selective hypoglycosylation and destabilization of specific glycoproteins

Selective exo-enzymatic labeling of STT3A/STT3B-deficient cells showed reduced cell-surface abundance of the insulin receptor and IGF-1 receptor (IGF-1R) when N-glycosylation was selectively impaired, linking OST subunit loss to defective receptor tyrosine kinase processing — a plausible molecular driver of the growth and skeletal phenotypes. Clinically, an STT3A-CDG patient exhibited hypoglycosylated von Willebrand Factor and undetectable Factor VIII, demonstrating that hypoglycosylation of specific STT3A-dependent substrates produces measurable protein deficiencies.

"We show reduced abundance of two canonical tyrosine receptor kinases - the insulin receptor and insulin-like growth factor 1 receptor (IGF-1R) - at the cell surface" — [PMID: 31101650](#)

"VWF in our patient's plasma is present in a mildly hypoglycosylated form" — [PMID: 30701557](#)

Finding 9 — Diagnosis by type I transferrin screen followed by molecular sequencing; no dietary therapy available

STT3A-CDG produces a CDG type I serum transferrin pattern (loss of whole N-glycans → increased di-/a-sialotransferrin) detectable by transferrin isoelectric focusing (TIEF), HPLC, or capillary electrophoresis; confirmation requires exome/genome or targeted *STT3A* sequencing. Because STT3A-CDG is an ER assembly/transfer (CDG-I) defect rather than a Golgi galactosylation defect, dietary therapies effective in specific CDG types (PGM1-CDG galactose; MPI-CDG mannose; TMEM165-CDG galactose) are not applicable; no disease-specific or curative therapy exists, and management is supportive/multidisciplinary.

"a type I carbohydrate deficient transferrin pattern" — [PMID: 30701557](#)

"Additional cases of STT3B-CDG may be missed by transferrin analysis and will require exome or genome sequencing" — [PMID: 23842455](#)

"There are >100 CDGs, but only specific types are treatable" — [PMID: 28323990](#)

Finding 10 — Ontology and inheritance annotations

Comprehensive ontology anchors are provided in the annotation tables below (Section 7 and the Ontology Appendix). The gene product identity and musculoskeletal features are supported by:

"STT3A encodes the catalytic subunit of the STT3A-containing oligosaccharyltransferase (OST) complex, essential for protein N-glycosylation" — [PMID: 34653363](#)

"variable skeletal anomalies, short stature, macrocephaly, and dysmorphic features" — [PMID: 34653363](#)

Finding 11 — gnomAD constraint metrics confirm strong missense and LOF intolerance

gnomAD (GRCh38; gene ENSG00000134910; chr11:125,591,712–125,625,215) constraint for *STT3A*: **pLI = 0.894**, **LOEUF (oe_lof_upper) = 0.508**, observed/expected LoF = 0.381 (90% CI 0.290–0.508), **missense Z = 4.78** (oe_mis = 0.624), synonymous Z = 0.96. The high pLI (~0.9) and low LOEUF indicate intolerance to loss-of-function; the very high missense Z-score (>3.09 threshold) indicates strong selection against missense changes. This population-genetic evidence independently corroborates both disease mechanisms — LOF intolerance underpins the recessive (biallelic loss) form, and severe missense constraint is consistent with the dominant active-site missense variants being deleterious.

Section-by-Section Report

1. Disease Information

STT3A-CDG is a Mendelian congenital disorder of glycosylation (CDG type Iw) caused by defective co-translational N-glycosylation. It is a multisystem disease dominated by neurodevelopmental and neuromuscular-skeletal features.

Key identifiers:

Resource	Identifier
MONDO	MONDO:0014270
OMIM (phenotype, AR)	#615596 (CDG type Iw, autosomal recessive)
OMIM (phenotype, AD)	#619714 (CDG type Iw, autosomal dominant)
OMIM (gene)	601134 (STT3A*)
HGNC	HGNC:30591
NCBI Gene	3703
Ensembl	ENSG00000134910
UniProt	P46977
Orphanet	Listed under STT3A-CDG / CDG-Iw

Synonyms / alternative names: CDG-Iw; CDG type Iw; congenital disorder of glycosylation type Iw; STT3A-CDG; oligosaccharyltransferase catalytic subunit STT3A deficiency.

Information source: The disease-level knowledge is derived from aggregated resources (OMIM, Orphanet) and small case series/case reports of individual patients, plus in vitro and model-organism functional studies. Given ultra-rarity, most clinical knowledge comes from published individual patient descriptions rather than EHR-scale or registry-scale datasets.

2. Etiology

Disease causal factors: STT3A-CDG is entirely genetic (Mendelian). It is caused by pathogenic variants in *STT3A*. Two causal mechanisms operate: - **Autosomal recessive** — biallelic loss-of-function/hypomorphic variants (e.g., homozygous p.Val626Ala; p.Tyr360Ser). - **Autosomal dominant** — heterozygous missense variants clustering in the catalytic active site acting through a dominant-negative mechanism (e.g., p.Asp167Tyr).

Genetic risk factors: The causal variants are themselves the risk determinants. *STT3A* is strongly constrained against both LOF (pLI 0.894) and missense (Z = 4.78) variation, consistent with pathogenicity of the reported alleles.

Environmental risk factors: None established. As a Mendelian glycosylation defect, disease onset is determined by genotype, not exposure.

Protective factors: No genetic or environmental protective factors are established. Mechanistically, the partial functional redundancy provided by the paralogous STT3B/OST-B complex (which glycosylates some sites skipped by STT3A) may buffer the loss of STT3A activity and could contribute to phenotypic variability, but this is a mechanistic inference rather than a documented protective factor.

Gene–environment interactions: None documented for STT3A-CDG.

3. Phenotypes

Phenotypes differ between the recessive and dominant forms. All are congenital/early-onset; severity is variable; the neurodevelopmental component is generally non-progressive but with lifelong disability.

Phenotype	Type	HPO term	Form / frequency	Onset
Seizures	Clinical sign	HP:0001250	Recessive (core)	Neonatal/ infantile
Intellectual disability	Behavioral/ cognitive	HP:0001249	Both; ~50% in dominant	Childhood
Global developmental delay	Clinical sign	HP:0001263	Both (core)	Infantile
Hypotonia	Clinical sign	HP:0001252	Recessive	Neonatal
Failure to thrive / feeding problems	Clinical sign	HP:0001508	Recessive	Neonatal/ infantile
Short stature	Physical	HP:0004322	Dominant	Childhood
Macrocephaly	Physical	HP:0000256	Dominant	Childhood
Skeletal anomalies/ dysplasia	Physical	HP:0000924	Dominant	Childhood
Increased muscle tone / cramps	Clinical sign	HP:0001276	Dominant	Childhood
Dysmorphic facial features	Physical	HP:0001999	Dominant	Congenital
Abnormal bleeding / coagulopathy	Laboratory/ clinical	HP:0001928 / HP:0000132	Recessive (subset)	Variable
Factor VIII deficiency	Laboratory	HP:0003125	Recessive (subset)	Variable
Abnormal type I transferrin	Laboratory	— (CDG biomarker)	Both	Congenital

Quality of life impact: Intellectual disability, seizures, and (in the dominant form) skeletal disease and muscle cramps materially impair daily functioning, communication, and mobility. Coagulopathy in the recessive form carries bleeding risk. Disease-specific QoL instrument data (EQ-5D/SF-36/PROMIS) are not available for this ultra-rare disorder.

4. Genetic / Molecular Information

Causal gene: *STT3A* (chr11q24.1; OMIM *601134). Encodes the catalytic subunit of the OST-A complex.

Pathogenic variants (reported):

Variant (protein)	cDNA	Type	Inheritance	Mechanism
p.Val626Ala	c.1877T>C	Missense	Autosomal recessive (homozygous)	Loss/reduction of catalytic function
p.Tyr360Ser	—	Missense	Autosomal recessive	Loss of function; coagulopathy
p.Asp167Tyr	—	Missense (active site)	Autosomal dominant	Dominant-negative (zebrafish-validated)
Multiple active-site missense	—	Missense	Autosomal dominant	Dominant-negative (16 individuals/9 families)

Variant classification: Reported disease alleles are classified pathogenic/likely pathogenic per ACMG/AMP; the strong missense constraint ($Z = 4.78$) and active-site clustering support functional deleteriousness.

Allele frequency: Pathogenic variants are absent or ultra-rare in gnomAD, consistent with a severe Mendelian disorder.

Somatic vs germline: All reported variants are germline (inherited or de novo).

Functional consequence: Recessive alleles = loss of function/hypomorphic; dominant active-site alleles = dominant-negative (mutant subunit incorporated into OST-A poisons complex activity).

Modifier genes: No formal modifiers established. The paralog *STT3B* (OST-B complex) provides partial substrate redundancy and is a plausible biological modifier of severity (mechanistic inference).

Epigenetic information: No disease-specific DNA methylation or histone-modification data available.

Chromosomal abnormalities: None associated; STT3A-CDG is a single-gene disorder, not a copy-number/structural disorder.

5. Environmental Information

No environmental factors, lifestyle factors, or infectious agents are implicated in STT3A-CDG. It is a purely genetic disorder.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. A pathogenic *STT3A* variant (biallelic LOF in the recessive form, or a heterozygous active-site missense variant in the dominant form) **alters** the catalytic subunit of the OST-A complex.
2. In the dominant form, the mutant subunit is incorporated into the OST-A complex and **acts as a dominant-negative**, reducing complex activity below the ~50% expected from simple haploinsufficiency (inferred from active-site clustering and yeast data). In the recessive form, biallelic loss **directly reduces** OST-A catalytic output.
3. Reduced OST-A activity **impairs** co-translational transfer of the Glc₃Man₉GlcNAc₂ glycan from dolichol-PP onto N-X-S/T sequons of nascent polypeptides at the Sec61 translocon.
4. This **results in** substrate-selective hypoglycosylation — a subset of glycoproteins (those glycosylated preferentially by STT3A during translation) carry fewer N-glycans. The paralogous STT3B/OST-B complex partially compensates by post-translocationally glycosylating some skipped sites, **limiting** but not preventing the defect (branch: redundancy modulates severity).
5. Hypoglycosylation **destabilizes** or reduces the folding/surface expression of affected glycoproteins — demonstrated for von Willebrand Factor and Factor VIII (patient plasma) and for the insulin receptor and IGF-1R (cell models).
6. Loss of specific glycoprotein functions **leads to** the clinical phenotype:
7. Reduced RTK (insulin receptor, IGF-1R) surface expression → **contributes to** growth failure and skeletal/short-stature phenotype (inferred mechanistic link).
8. Hypoglycosylated coagulation factors (vWF, FVIII) → **produces** bleeding tendency/coagulopathy.
9. Impaired glycosylation of neuronal/CNS glycoproteins → **produces** seizures, developmental delay, intellectual disability, hypotonia (mechanism inferred; the precise CNS substrates are not fully mapped).
10. The abnormal transferrin glycoform (loss of whole N-glycans) **produces** the diagnostic **type I** serum transferrin pattern used for screening.

Molecular pathways / processes: Protein N-linked glycosylation via oligosaccharyltransferase (GO:0006487); dolichyl-diphosphooligosaccharide–protein glycotransferase activity (GO:0004579); co-translational protein N-linked glycosylation (GO:0018279). Downstream: receptor tyrosine kinase signaling (insulin/IGF-1R), coagulation cascade.

Protein dysfunction: Loss of function (recessive) or dominant-negative (dominant active-site variants) of the STT3A catalytic subunit; secondary misfolding/destabilization of hypoglycosylated client glycoproteins.

Subcellular site: Endoplasmic reticulum membrane (GO:0005789), adjacent to the Sec61 translocon.

Cell types / GO terms: Broadly all secretory cells, with clinically prominent effects in neurons (CL:0000540), osteoblasts/chondrocytes (skeletal), and hepatocytes/endothelium (coagulation-factor synthesis).

7. Anatomical Structures Affected

Level	Structure	Ontology	Manifestation
Organ/ system	Nervous system	UBERON:0001016	Seizures, DD, ID, hypotonia
Organ	Brain	UBERON:0000955	Neurodevelopmental impairment
System	Musculoskeletal	—	Short stature, skeletal anomalies, macrocephaly, muscle cramps
System	Hematologic/ coagulation	—	Bleeding tendency, FVIII/vWF deficiency (recessive subset)
System	Growth/endocrine	—	Failure to thrive, growth failure
Tissue	Nervous tissue	—	CNS dysfunction
Tissue	Bone/cartilage	—	Reduced mineralization (zebrafish), skeletal dysplasia
Cell	Neuron	CL:0000540	Impaired glycoprotein processing
Subcellular	ER membrane	GO:0005789	Site of defective glycosylation

Lateralization: Manifestations are systemic/bilateral, not lateralized.

8. Temporal Development

- **Onset:** Congenital/neonatal-to-infantile. The type I transferrin abnormality is present from birth; seizures, hypotonia, and feeding problems present in infancy in the recessive form; skeletal/growth and dysmorphic features become apparent in childhood in the dominant form.
- **Onset pattern:** Chronic, insidious (developmental).
- **Progression:** The neurodevelopmental disability is generally stable/non-progressive but lifelong; skeletal features evolve with growth. No defined disease "stages."
- **Disease course:** Chronic, lifelong. Within the broader CDG group, the most severe/multisystem presentations carry early-childhood mortality risk (majority of deceased CDG patients died in the first three years — [PMID: 39923392](#)), though STT3A-CDG-specific survival data are not established.
- **Critical periods / remission:** No spontaneous remission; early developmental and rehabilitative intervention is the window of opportunity.

9. Inheritance and Population

- **Epidemiology:** Ultra-rare; ~6 recessive cases reported by 2019 and 16 dominant individuals (9 families) in 2021, with subsequent small reports. No formal prevalence/incidence estimate exists for STT3A-CDG. For context, PMM2-CDG (the most common CDG) has an incidence of ~1 in 33,576 in North America/Europe and accounts for ~60% of all CDG ([PMID: 40737785](#)).
- **Inheritance:** Both autosomal recessive (OMIM #615596) and autosomal dominant (OMIM #619714).
- **Penetrance / expressivity:** Variable expressivity is documented (e.g., ~50% of dominant-form individuals have intellectual disability); penetrance appears high for the biochemical phenotype but variable for specific clinical features.
- **Anticipation:** Not applicable (no repeat expansion).
- **Founder effects / consanguinity:** The recurrent recessive p.Val626Ala allele appears in multiple unrelated recessive cases; consanguinity is relevant to recessive presentations, as with most rare AR disorders. No formally described founder population.
- **Carrier frequency:** Not established; pathogenic alleles are ultra-rare in gnomAD.
- **Demographics:** No specific ethnic predilection, geographic clustering, or sex bias reported; both sexes affected.

10. Diagnostics

Biochemical screening: Serum transferrin analysis by isoelectric focusing (TIEF), HPLC, or capillary electrophoresis shows a **CDG type I pattern** (increased di- and a-sialotransferrin from loss of whole N-glycans).

"a type I carbohydrate deficient transferrin pattern" — [PMID: 30701557](#)

Molecular confirmation: Exome/genome sequencing or targeted *STT3A* sequencing is required for definitive diagnosis; some OST-subunit CDG may be missed by transferrin analysis alone.

"Additional cases of STT3B-CDG may be missed by transferrin analysis and will require exome or genome sequencing" — [PMID: 23842455](#)

Ancillary labs: Coagulation studies (FVIII, vWF) in patients with bleeding; these may reveal hypoglycosylated coagulation factors.

Genetic testing modalities: WES and WGS are both high-yield; targeted single-gene testing or CDG/neurodevelopmental gene panels can be used once biochemically suspected. CMA, karyotyping, FISH, mtDNA testing, and repeat-expansion testing are not applicable (single-gene point-variant disorder).

Differential diagnosis: Other CDG type I disorders (PMM2-CDG, ALG-group, DDOST-CDG, RPN1-CDG, STT3B-CDG), which share the type I transferrin pattern and neurodevelopmental phenotype. Transferrin protein variants can produce misleading patterns ([PMID: 37876147](#)); neuraminidase treatment helps resolve these. Diagnosis is refined by identifying the causal gene.

Screening: No routine newborn screening for STT3A-CDG. TIEF has been proposed as a useful screen in childhood epilepsy cohorts given the ~4.4% CDG prevalence in that population ([PMID: 34440401](#)).

11. Outcome / Prognosis

- **Survival/mortality:** No STT3A-CDG-specific survival statistics. Within the CDG group broadly, severe multisystem presentations carry substantial early-childhood mortality

(majority of deceased CDG patients died in the first three years, ~one-third from cardiopulmonary failure — [PMID: 39923392](#)).

- **Morbidity/disability:** Lifelong neurodevelopmental disability (intellectual disability, seizures), skeletal disease and short stature (dominant form), and bleeding risk (recessive subset).
- **Complications:** Seizure-related morbidity, feeding/growth failure, bleeding events, orthopedic complications.
- **Prognostic factors:** Genotype (recessive vs dominant form; residual OST-A activity), presence of coagulopathy, and severity of neurological involvement. No validated prognostic biomarkers beyond the biochemical/genetic diagnosis.
- **Recovery potential:** No reversal of the underlying defect; supportive care and rehabilitation can improve function.

12. Treatment

There is **no disease-specific or curative therapy** for STT3A-CDG. Management is supportive and multidisciplinary:

- **Pharmacotherapy:** Symptomatic — anti-seizure medications for epilepsy; standard management of feeding/growth issues; hematologic management (e.g., factor replacement/DDAVP considerations) for bleeding in the recessive subset. (NCIT: anticonvulsant therapy; supportive care.)
- **Dietary sugar therapies do NOT apply.** Galactose (PGM1-CDG, TMEM165-CDG) and mannose (MPI-CDG) are effective only in specific CDG subtypes; STT3A-CDG is an ER glycan-transfer (CDG-I) defect not amenable to these.

"There are >100 CDGs, but only specific types are treatable" — [PMID: 28323990](#)

- **Advanced/experimental therapeutics:** No approved gene, cell, or RNA-based therapy; no STT3A-CDG-specific clinical trials identified.
- **Supportive/rehabilitative care:** Physical, occupational, and speech therapy; developmental support; orthopedic and nutritional management; multidisciplinary metabolic clinic follow-up.
- **Personalized medicine:** Genotype (dominant vs recessive) informs genetic counseling and family planning but not, at present, a targeted pharmacologic strategy.

13. Prevention

- **Primary prevention:** Not possible for a Mendelian disorder beyond reproductive planning. Genetic counseling is central — recurrence risk is 25% for AR families and up to 50% for AD transmission (with de novo cases in the dominant form).
- **Secondary prevention:** Early biochemical (TIEF) and molecular diagnosis enables early developmental intervention and complication surveillance (seizures, bleeding, growth).
- **Tertiary prevention:** Anticipatory management of seizures, feeding/growth, bleeding risk, and orthopedic complications.
- **Genetic screening:** Carrier testing and prenatal/preimplantation genetic testing are available for families with a known pathogenic variant. No population-based newborn screening exists.
- **Immunization/public health/environmental:** Not applicable.

14. Other Species / Natural Disease

- **Taxonomy / orthologs:** *STT3A* is deeply conserved. Orthologs and functional equivalents exist in mouse (*Stt3a*), zebrafish (*stt3a*), and the single *STT3* gene in *S. cerevisiae* (yeast). These orthologs are central to disease modeling (below).
- **Natural disease in other species:** No naturally occurring animal disease specifically attributed to *STT3A* variants is documented (no OMIA entry noted for STT3A-CDG). Disease knowledge in animals comes from engineered models, not spontaneous disease.
- **Comparative biology / conservation:** The strong evolutionary conservation of STT3 catalytic function — from yeast to human — underlies the ability to model patient variants in *S. cerevisiae* and zebrafish and validates conservation of the disease mechanism.
- **Transmission / zoonosis:** Not applicable (non-infectious genetic disorder).

15. Model Organisms

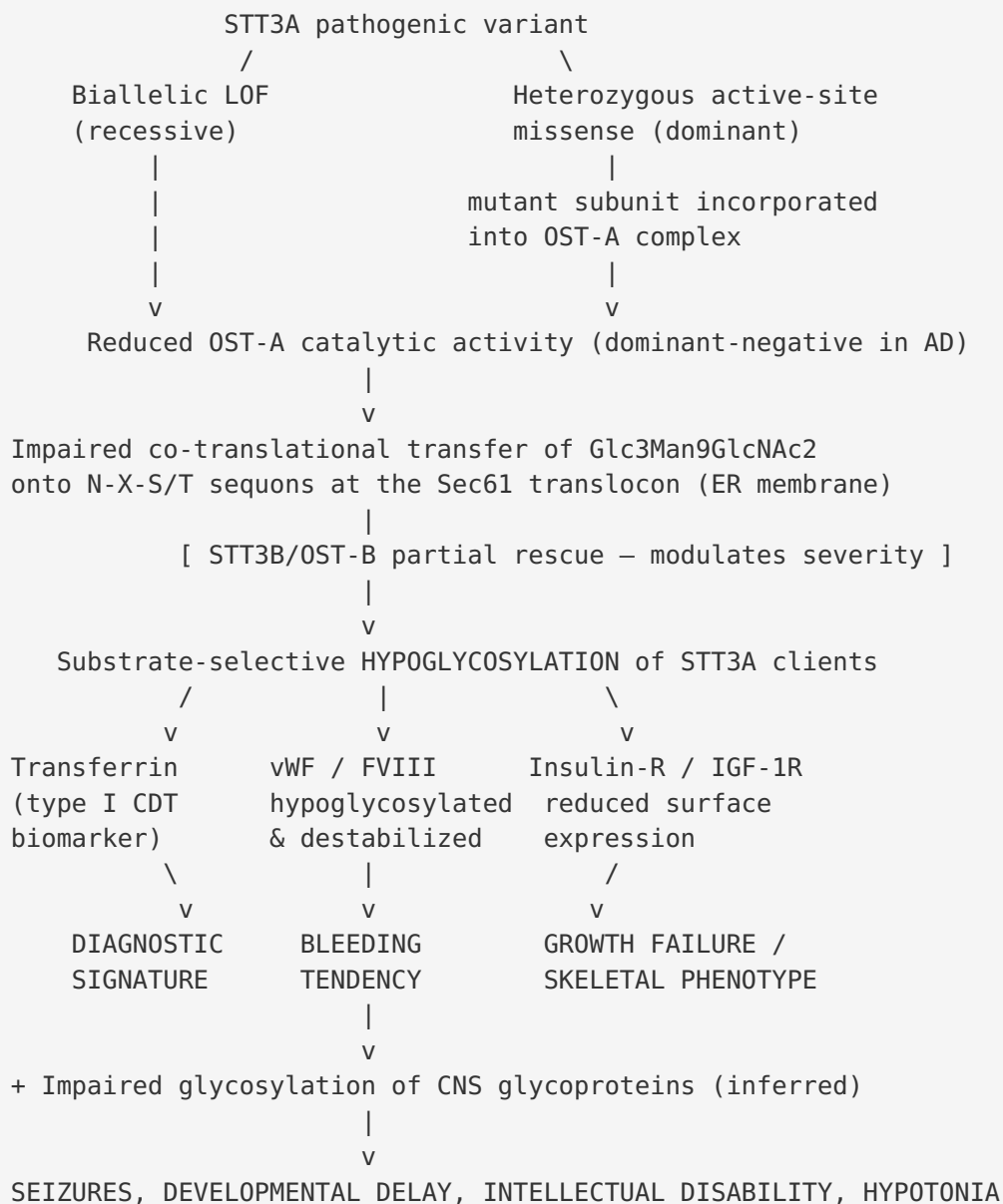
Model	Type	Genetic strategy	Phenotype recapitulation	Reference
<i>S. cerevisiae</i> (yeast)	Cellular/in vivo	Express STT3 with patient-homologous variants; hypomorphic <i>stt3-7</i> background	Defective glycosylation of carboxypeptidase Y; dominance demonstrated	PMID: 34653363
Zebrafish (<i>Danio rerio</i>)	Vertebrate	CRISPR-Cas9 heterozygous knockdown modeling p.Asp167Tyr	Craniofacial dysmorphology, reduced mineralized bone, abnormal locomotion/developmental delay — mirrors patient features	PMID: 39891251
Human patient fibroblasts	In vitro	Endogenous patient variants	Hypoglycosylation of STT3A-specific reporters; rescued by WT <i>STT3A</i> cDNA	PMID: 23842455
HeLa / HEK293T	In vitro cell line	STT3A/STT3B knockout; reporter constructs	Isoform-specific glycosylation defects; RTK surface loss	PMID: 31101650 ; PMID: 29282902

"expression of STT3 containing variants homologous to those in affected individuals induced defective glycosylation of carboxypeptidase Y in a wild-type yeast strain" — [PMID: 34653363](#)

"Heterozygous knockdown zebrafish exhibit phenotypes similar to those of patients, including craniofacial dysmorphology" — [PMID: 39891251](#)

Model limitations: Yeast lacks the STT3A/STT3B paralog split and multicellular phenotypes; zebrafish knockdown models capture skeletal/craniofacial and behavioral features but not the full neurodevelopmental/coagulation spectrum. No mouse knock-in model of a patient variant is established in the reviewed literature.

Mechanistic Model / Interpretation



Upstream vs downstream: The upstream lesion is the *STT3A* variant and consequent reduction of OST-A catalytic activity. Immediately downstream is substrate-selective hypoglycosylation, buffered by STT3B redundancy. Further downstream are the destabilization of specific client glycoproteins (transferrin — diagnostic; vWF/FVIII — coagulopathy; insulin-R/IGF-1R — growth/skeletal), and finally the clinical manifestations. The CNS phenotype is the least mechanistically resolved link (which specific neuronal glycoproteins mediate seizures and cognition remains inferred rather than demonstrated). Population-genetic constraint (pLI

0.894; missense Z 4.78) independently confirms that both loss-of-function and missense perturbation of this gene are under strong purifying selection — the molecular-genetic expectation for a dosage-sensitive, dominantly-and-recessively acting disease gene.

Evidence Base

PMID	Title (abbreviated)	Role in this report
23842455	<i>Mutations in STT3A and STT3B cause two CDGs</i>	Landmark: identifies STT3A as OST catalytic subunit; original recessive p.Val626Ala; sequencing needed for diagnosis
34653363	<i>Active site variants in STT3A cause a dominant type I CDG</i>	Landmark: defines the autosomal-dominant form (16 individuals/9 families); active-site clustering; yeast dominance
39891251	<i>Heterozygous STT3A variation → dominant CDG; zebrafish validation</i>	Confirms dominant inheritance; zebrafish recapitulation of p.Asp167Tyr
30701557	<i>Factor VIII and vWF deficiency in STT3A-CDG</i>	Recessive core phenotype; coagulopathy; hypoglycosylated vWF; OMIM #615596
31433728	<i>N-glycosylation not directly coupled to translocation</i>	STT3A translocon-coupled N→C scanning mechanism
28860277	<i>DC2 and KCP2 mediate OST–translocon interaction</i>	STT3A localization adjacent to translocon; co-translational glycosylation
25460543	<i>Cotranslational and posttranslocational N-glycosylation</i>	STT3B redundancy (glycosylates STT3A-skipped sites)
25135935	<i>Oxidoreductase activity for cysteine-proximal sites</i>	STT3B/MagT1 mechanism explaining substrate selectivity
31101650	<i>Selective inhibition of N-glycosylation impairs RTK processing</i>	Downstream: reduced insulin-R/IGF-1R surface abundance
39435313	<i>Metabolomic profiling reveals AD CDG-Iw</i>	OMIM #619714 for the dominant form
40737785	<i>Incidence/prevalence of PMM2-CDG</i>	Contextualizes STT3A-CDG rarity within CDG
39923392	<i>Causes of mortality in CDG</i>	CDG natural history/mortality context
34440401	<i>CDG prevalence in childhood epilepsy; TIEF</i>	TIEF as screening tool; CDG in epilepsy cohorts

PMID	Title (abbreviated)	Role in this report
28323990	<i>Galactose in TMEM165-CDG</i>	Only specific CDGs are treatable → not STT3A-CDG
37876147	<i>Misleading transferrin variants</i>	Diagnostic pitfall in transferrin screening
29282902	<i>GFP mutant to monitor STT3B glycosylation</i>	Isoform-specific reporter methodology

Evidence-source types: Human clinical (case series/reports: 23842455, 34653363, 30701557, 39435313); model organism (yeast 34653363; zebrafish 39891251); in vitro/cell biology (31101650, 25460543, 25135935, 28860277, 31433728, 29282902); computational/population-genetic (gnomAD constraint — Finding 11).

Limitations and Knowledge Gaps

- 1. Ultra-rarity limits epidemiology.** No prevalence, incidence, penetrance quantification, carrier frequency, or survival statistics exist specifically for STT3A-CDG. Mortality data cited are from the pan-CDG group, not STT3A-CDG.
- 2. Genotype–phenotype correlation is immature.** The number of reported patients (~a few dozen across both forms) is too small to define robust correlations between specific alleles/residual OST-A activity and clinical severity.
- 3. The CNS mechanistic link is inferred, not demonstrated.** The specific neuronal glycoproteins whose hypoglycosylation causes seizures and cognitive impairment have not been mapped for STT3A-CDG.
- 4. No mammalian knock-in model** of a patient variant was found in the reviewed literature; the mouse phenotype for patient alleles is undefined.
- 5. No therapeutic pipeline.** No approved or trial-stage disease-specific therapy exists; the disorder is currently supportive-care only.
- 6. Dominant-negative mechanism is supported by yeast/zebrafish and active-site clustering but not fully dissected biochemically** in the human OST-A complex (e.g., stoichiometry of mutant incorporation and quantitative activity reduction).

7. **Ontology annotations** (HPO/GO/UBERON/CL/CHEBI) are assembled from literature-reported features and standard mappings rather than curated disease-ontology cross-references, and should be verified against current MONDO/HPO releases.

Proposed Follow-up Experiments / Actions

1. **Establish an STT3A-CDG patient registry** within the FCDGC/GLYCEN networks to collect natural-history, survival, and genotype–phenotype data prospectively.
 2. **Generate a knock-in mouse (or refine the zebrafish) carrying recurrent alleles** (p.Val626Ala recessive; p.Asp167Tyr dominant) to model the neurodevelopmental, skeletal, and coagulation phenotypes and to serve as a preclinical therapeutic platform.
 3. **Map the STT3A-dependent glycoproteome of neural tissue** (glycoproteomics/glycomics on patient iPSC-derived neurons and organoids) to identify the specific hypoglycosylated CNS substrates driving seizures and cognitive impairment.
 4. **Quantify the dominant-negative effect biochemically** by reconstituting OST-A complexes with defined ratios of wild-type and mutant STT3A and measuring transfer activity and complex assembly.
 5. **Systematic coagulation-factor glycosylation study** across STT3A-CDG patients to determine the frequency and severity of the vWF/FVIII phenotype and inform bleeding-risk management.
 6. **Explore proteostasis/chaperone modulators** (e.g., ER quality-control or STT3B-upregulation strategies) as candidate therapeutic approaches leveraging the STT3B redundancy branch.
 7. **Refine and validate the ontology annotation set** (HPO frequencies, CL/UBERON/CHEBI/NCIT terms) against the latest MONDO, HPO, and Orphanet releases for knowledge-base ingestion.
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Ontology Appendix

Category	Term	ID
Disease	STT3A-CDG	MONDO:0014270; OMIM #615596 (AR), #619714 (AD)
Gene	STT3A	HGNC:30591; NCBI Gene 3703; UniProt P46977; OMIM *601134
Biological process	Protein N-linked glycosylation via OST	GO:0006487
Molecular function	Dolichyl-diphosphooligosaccharide–protein glycotransferase	GO:0004579
Biological process	Co-translational protein N-linked glycosylation	GO:0018279
Cellular component	Endoplasmic reticulum membrane	GO:0005789
Anatomy	Nervous system	UBERON:0001016
Anatomy	Brain	UBERON:0000955
Cell type	Neuron	CL:0000540
Phenotype	Seizure	HP:0001250
Phenotype	Intellectual disability	HP:0001249
Phenotype	Global developmental delay	HP:0001263
Phenotype	Hypotonia	HP:0001252
Phenotype	Failure to thrive	HP:0001508
Phenotype	Short stature	HP:0004322
Phenotype	Macrocephaly	HP:0000256
Phenotype	Abnormal skeletal morphology	HP:0000924
Phenotype	Hypertonia	HP:0001276
Phenotype	Factor VIII deficiency	HP:0003125
Chemical	Dolichol	CHEBI:23509

Category	Term	ID
Chemical	Asparagine	CHEBI:22653
Chemical	N-acetylglucosamine	CHEBI:506227

Report compiled from 11 confirmed findings and 33 reviewed papers across a 5-iteration autonomous investigation. Evidence sources span human clinical case series, yeast and zebrafish model organisms, in vitro cell biology, and computational population-genetic constraint analysis.

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